

SUMIT YADAV

Lead Software Engineer. Building intelligent, scalable software systems

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PROFESSIONAL SUMMARY

Lead Software Engineer with **7+ years** of experience architecting and delivering intelligent, scalable software systems across cloud-native, distributed, and AI-powered applications. Currently providing technical leadership across multiple enterprise full-stack and microservices initiatives involving **30+** engineers at **Accenture**. Passionate about designing reliable systems, simplifying complex engineering challenges, and building software that scales. Author of a technical blog focused on distributed systems, system design, and AI engineering.

TECHNICAL SKILLS

Full-Stack Engineering

Node.js, TypeScript, React, Next.js, GraphQL, Tailwind CSS

AI Engineering

LLMs, RAG Pipelines, Prompt Engineering, Google Gemini, Vercel AI SDK, Pinecone, Vector Embeddings

Databases & Caching

Redis (Upstash), PostgreSQL, DynamoDB, MongoDB, Pinecone (Vector DB)

Cloud & Infrastructure

AWS (CloudFront, Lambda, DynamoDB, SQS, SNS, S3), Azure (Functions, App Service, DevOps, Storage), Docker, Kubernetes, Nginx, Vercel

Architecture & Systems

Distributed Systems, System Design, Microservices, Event-Driven Architecture, Kafka, REST APIs, WebSockets, Server-Sent Events

Leadership & Engineering Practices

Technical Leadership, Technical Mentorship, Architecture Reviews, Code Reviews, CI/CD, Performance Optimisation, Agile/Scrum

PROFESSIONAL EXPERIENCE

Lead Software Engineer · Accenture Solutions Private Limited

Mumbai, India | Dec 2021 – Present

- Provided technical leadership, architectural guidance, and engineering mentorship across multiple full-stack and microservices initiatives involving 30+ engineers, driving engineering standards, code quality, and scalable system design.
- Designed and delivered Node.js backend modules and REST APIs for high-traffic enterprise applications, achieving 20–30% API performance improvement through targeted caching strategies and query optimisation
- Architected and implemented Azure-based cloud solutions including Azure Functions, App Service, and Storage Accounts, integrated into CI/CD workflows via Azure DevOps pipelines
- Partnered with solution architects to define service boundaries, deployment strategies, and communication contracts for large-scale microservice ecosystems.
- Designed and delivered production-grade Retrieval-Augmented Generation (RAG) pipelines integrating LLMs, vector search, and semantic retrieval using Next.js, Google Gemini, Pinecone, and the Vercel AI SDK.
- Mentored engineers in scalable design patterns, cloud-native development, security best practices, and TypeScript — consistently raising team technical standards
- **Recognition:** *Care Deeply For All Our People Award (Feb 2024)* · *Incredible Ownership Award (Sep 2023)*

Senior Software Engineer · Infosys Limited

Pune, India | Sep 2021 – Dec 2021

- Delivered full-stack features across modern JavaScript and Node.js applications while collaborating with cross-functional teams in an enterprise environment.

Software Engineer · Accenture Solutions Private Limited

Mumbai, India | Jul 2020 – Sep 2021

- Built enterprise React and TypeScript applications focused on performance, maintainability, and reusable component architecture. Contributed to component libraries, responsive design systems initiatives.

Assistant System Engineer · Tata Consultancy Services

Mumbai, India | Dec 2018 – Dec 2019

- Built and maintained enterprise web applications using JavaScript and Node.js while contributing across frontend and backend services.
- Recognition:** *Kudos Award for Excellent Performance (Feb 2019)*

ENGINEERING PROJECTS

AlgoLens — Interactive Algorithm Visualizer · [Live Demo](#) · [Source Code](#)

Next.js · TypeScript · Vanilla CSS · React · Vercel

- Designed and built 25+ interactive step-by-step algorithm and system design visualizers from scratch — zero external animation libraries — covering sorting, searching, graphs, trees, dynamic programming, and distributed systems flows including LRU Cache and Consistent Hashing
- Architected a universal AnimationStep contract decoupling algorithm logic from the UI — every visualizer is a pure function producing steps consumed by a shared playback engine, making new algorithms addable in 3 files

AI-Powered RAG Assistant · [Live Demo](#) · [Source Code](#)

Next.js · Google Gemini · Pinecone · Vercel AI SDK · TypeScript · Upstash Redis

- End-to-end RAG pipeline — PDF ingestion, text chunking, Gemini vector embeddings, Pinecone semantic search, and real-time streaming responses with source citations
- Production-hardened with embedding cache, rate limiting (Upstash sliding window), and mock dev mode to control API costs

Real-Time Notification Engine · [Live Demo](#) · [Source Code](#)

Next.js · SSE · Upstash Redis · QStash · TypeScript · Vercel Edge

- Event-driven notification system with SSE delivery, Redis persistence, and QStash fan-out across In-App, Email, and SMS channels with auto-reconnection logic
- Designed around event-driven architecture with asynchronous fan-out workers and eventual consistency.

Distributed URL Shortener · [Live Demo](#) · [Source Code](#)

Next.js · Upstash Redis · Vercel Edge Functions · TypeScript

- Designed around AP principles of the CAP theorem with Redis-backed lookups and globally distributed edge redirects.

TECHNICAL WRITING

Author of a technical blog sumityadav-dev.vercel.app — covering distributed systems, AI engineering, and modern software architecture with deep dives into SAGA patterns, RAG systems, message queues, CAP theorem, consistent hashing, and large-scale system design.

EDUCATION

B.Tech in Computer Science and Engineering · Manipal University Jaipur

Jaipur, India | 2014 – 2018 | CGPA: 8.03 / 10

Portfolio: ysumit99.github.io | Blog: sumityadav-dev.vercel.app